

Angle Ranks of Abelian Varieties

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Slides available at <https://dmzb.github.io/>

$$\Gamma := \langle \alpha_1, \dots, \alpha_n \rangle \subset \overline{\mathbb{Q}}^*$$
$$\alpha_1^{A_1} \cdots \alpha_n^{A_n} = 1$$

Let $f(x) \in \mathbb{Z}[x]$ be monic, with nonzero roots $\alpha_1, \dots, \alpha_n$.

Definition

$$\begin{aligned}\overline{\mathbb{Q}}^* \supset \Gamma &:= \langle \alpha_1, \dots, \alpha_n \rangle \cong \mathbb{Z}^r \oplus T & (\text{Note: } r \leq n) \\ S^1 \supset \Gamma' &:= \langle \theta_1, \dots, \theta_n \rangle \cong \mathbb{Z}^\delta \oplus T & (\alpha_j = r_j e^{i\theta_j})\end{aligned}$$

Example

$$\begin{aligned}\Phi_n(x) \text{ cyclotomic} &\Rightarrow \boxed{r = 0} & (\text{since } \alpha_i = \zeta) \\ x^n - b &\Rightarrow \boxed{r = 0 \text{ or } 1} & (\text{since } \alpha = \zeta b^{\frac{1}{n}})\end{aligned}$$

Lemma: If $n > 3$ and $\text{Gal } f \cong S_n$, then $\boxed{r = n \text{ or } n-1}$

Proof.

Suppose $\alpha_1^{A_1} \cdots \alpha_n^{A_n} = 1$.

Apply $\tau = (ij)$ and divide to get $\left(\frac{\alpha_i}{\alpha_j}\right)^{A_i - A_j} = 1$.

Then $\alpha_i = \zeta \alpha_j$. So $\# \text{Gal } f \leq n(n-1)$. □

Weil Polynomials. Let $q = p^n$.

Definition

A q -Weil Polynomial $f(x) \in \mathbb{Z}[x]$ is a **monic** polynomial of **even degree** such that **all roots have complex absolute value** $\sqrt{q}$.

Definition

A q -Weil number α is a root of a q -Weil polynomial.

Roots and slopes

$$\alpha_1, \overline{\alpha_1}, \dots, \alpha_g, \overline{\alpha_g} \in \overline{\mathbb{Z}}$$

$$\alpha_i \cdot \overline{\alpha_i} = q$$

$$v_q(\alpha_i) \in [0, 1] \cap \mathbb{Z} \left[\frac{1}{2g} \right]$$

Ranks

① $r \leq g + 1$

② $\delta = r - 1 \leq g$

Let $A/\mathbb{F}_q$ be an Abelian Variety of dimension g

Weil Conjectures.

- 1 Let $P_A(x)$ be the characteristic polynomial $P_A(x)$ of $F \in \text{End}_{\mathbb{Z}_\ell}(T_\ell A)$.
- 2 Then $P_A(x)$ is a q -Weil polynomial.
- 3 $\#A(\mathbb{F}_q^n) = (\alpha_1^n + \overline{\alpha}_1^n + \cdots + \alpha_g^n + \overline{\alpha}_g^n) - (\alpha_1^n \alpha_2^n + \cdots + \alpha_{g-1}^n \alpha_g^n) + \cdots + q^g$

“Linear algebra”

$$H^i(A) \cong \wedge^i H^1(A)$$

Theorem (Honda–Tate)

The map

$$\left\{ \begin{array}{l} \text{isogeny classes of} \\ \text{simple abelian varieties over } \mathbb{F}_q \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{conjugacy classes of} \\ q\text{-Weil numbers} \end{array} \right\}$$

$$\text{isogeny class of } A \longmapsto \text{the set of zeros of } P_A(x)$$

is a bijection.

Galois group of a Weil Polynomial $f(x)$

Roots

$$\alpha_1, \overline{\alpha_1}, \dots, \alpha_g, \overline{\alpha_g} \in \overline{\mathbb{Z}}$$

$$\alpha_i \cdot \overline{\alpha_i} = q$$

Definition

$$W_{2g} = \text{Aut}(\{1, \overline{1}\}, \{2, \overline{2}\}, \dots, \{g, \overline{g}\}) \subset S_{2g}$$

$$\begin{array}{ccccc} C & \hookrightarrow & G & \twoheadrightarrow & \overline{G} \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{F}_2^g & \hookrightarrow & W_{2g} & \twoheadrightarrow & S_g \end{array}$$

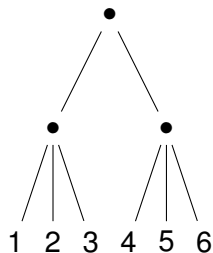
- 1 C is a **linear code**
- 2 C and $\dim_{\mathbb{F}_2} C$ are invariants of f (and A)

Theorem (Dupuy–Kedlaya–Zureick-Brown)

If $\#C > 2$, f is irreducible, and $G = \text{Gal } f(x)$ is primitive, then $\boxed{\delta = g}$.

Definition

- ① **Primitive** means “no blocks”.
- ② A “**Block**” is $B \subset \{\alpha_1, \overline{\alpha_1}, \dots, \alpha_g, \overline{\alpha_g}\}$ such that $1 < |B| < 2g$ and such that for all $\sigma \in G$, $S \cap \sigma(B) = B$ or $\emptyset$.



$\text{Aut } T \subset \{1, \dots, 6\}$ is transitive, but not primitive.
 $B = \{1, 2, 3\}$ is a block.

Theorem (Dupuy–Kedlaya–Zureick-Brown)

We construct a representation $V \in \text{Rep } G$ such that $\dim V = \delta$.

This recovers all previous work.

- 1 [Tankeev]: g prime $\Rightarrow \delta \in \{1, g-1, g\}$.
- 2 [Lenstra–Zarhin]: A “almost ordinary”^a $\Rightarrow \delta \in \{g-1, g\}$.
- 3 [DKZB]: If f is irreducible and $C = \mathbb{F}_2^g$, then $\delta = g$.

^aSlopes = $(0, \dots, 0, 1/2, 1/2, 1, \dots, 1)$

Database of Weil polynomials (Dupuy–Kedlaya–Roe–Vincent)

- 1 Many improved tools for fast computation
- 2 Disproof of Shparlinski’s conjecture (ordinary $\Rightarrow \delta = g$).

Why Bother?

- 1 The **Tate Conjecture** is true for all powers A (assuming $\delta = g$).
- 2 **Arithmetic Statistics**: distribution of $\#A(\mathbb{F}_q^n)$ (Arango-Piñeros–Bhadmidipati–Sankar)
- 3 **Inverse Galois** for Weil polynomials (Arango-Piñeros–Frengly–Vemulapalli)
- 4 More new cases of the **Tate Conjecture** (Arango-Piñeros–Frengly–Vemulapalli)

Observations

Conjugate

If $\prod \alpha_i^{A_i} = q^N$, then $\prod \overline{\alpha_i}^{A_i} = q^N$

Divide

Let $\beta_i = \alpha_i / \overline{\alpha_i}$, and divide to get $\prod \beta_i^{A_i} = 1$

Note: slopes shift

$v_q(\beta_i) \in [-1, 1]$ and $v_q(1/\beta_i) = -v_q(\beta_i)$

Linearize

Applying v_q gives $\sum A_i v_q(\beta_i) = 0$.

Let $\sigma \in \text{Gal } f$

If $\prod \beta_i^{A_i} = 1$, then $\prod \sigma(\beta_i)^{A_i} = 1$, so $\sum A_i v_q(\sigma(\beta_i)) = 0$.

Main point

Recap: for $\sigma \in \text{Gal } f$

$$\prod \beta_i^{A_i} = 1 \Rightarrow \boxed{\sum A_i v_q(\sigma(\beta_i)) \stackrel{*}{=} 0}$$

The converse holds [Zarhin, Zywina]

If $A_1, \dots, A_g \in \mathbb{Z}$ satisfy $*$ for all $\sigma \in \text{Gal } f$, then

$$\prod \beta_i^{A_i} = \zeta.$$

“Newton hyperplane” representation

$$\prod \beta_i^{A_i} = \zeta \Leftrightarrow \boxed{\sum A_i v_q(\sigma(\beta_i)) = 0} \text{ for all } \sigma \in \text{Gal } f.$$

$$M = (v_q(\sigma(\beta_j))) = \begin{pmatrix} v_q(\beta_1) & v_q(\beta_2) & \cdots & v_q(\beta_g) \\ v_q(\sigma(\beta_1)) & v_q(\sigma(\beta_2)) & \cdots & v_q(\sigma(\beta_g)) \\ \vdots & \vdots & \ddots & \vdots \\ v_q(\sigma'(\beta_1)) & v_q(\sigma'(\beta_2)) & \cdots & v_q(\sigma'(\beta_g)) \end{pmatrix}$$

Proposition (Let $\bar{A} = (A_1, \dots, A_g)$)

$$\prod \beta_i^{A_i} = \zeta \Leftrightarrow M \cdot \bar{A}^T = 0$$

$$\mathbb{Q}^g \supset V = \text{Span} \langle v_q(\sigma(\bar{\beta})) : \sigma \in \text{Gal } f \rangle$$

Corollary

$$\delta = \text{rank } M = \dim_{\mathbb{Q}} V.$$

(Example) Suppose that $\#C = 2$, and $\text{Gal } f \cong \langle \sigma = (123456) \rangle$

Re-index the roots so that σ acts as (123456) . Then both

$v_q(\bar{\beta}) = (1, 1, 1, -1, -1, -1)$ or $(1, -1, 1, -1, 1, -1)$ are possible.

In the first case, $\delta = 6$, and in the second case, $\delta = 2$

Matrices

$$\begin{pmatrix} 1 & 1 & 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 & 1 \end{pmatrix} \text{ vs. } \begin{pmatrix} 1 & -1 & 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 & -1 & 1 \end{pmatrix}$$

Remark

One can't "simultaneously choose" coordinates for both σ and v_q .

Sample Corollary

Suppose that $\#C = \mathbb{F}_2^g$ and f is irreducible. Then $\delta = g$.

Recall: $\beta_i = \alpha_i / \overline{\alpha_i}$.

Proof.

Note that for any $e_i \in C$,

$$e_i(\beta_j) = \beta_j^{(-1)^{\delta_{i,j}}}.$$

Thus

$$v_q(e_i(\beta_j)) = (-1)^{\delta_{i,j}} v_q(\beta_j).$$

Now, if $(a_1, \dots, a_j, \dots, a_g) \in V$, then $(a_1, \dots, -a_j, \dots, a_g) \in V$.

Thus,

$$(0, \dots, 0, 2a_j, 0, \dots, 0) \in V$$



Thank You!

$$\langle \alpha_1, \dots, \alpha_n \rangle \subset \overline{\mathbb{Q}}^*$$

$$\begin{pmatrix} v_q(\beta_1) & v_q(\beta_2) & \cdots & v_q(\beta_g) \\ v_q(\sigma(\beta_1)) & v_q(\sigma(\beta_2)) & \cdots & v_q(\sigma(\beta_g)) \\ \vdots & \vdots & \ddots & \vdots \\ v_q(\sigma'(\beta_1)) & v_q(\sigma'(\beta_2)) & \cdots & v_q(\sigma'(\beta_g)) \end{pmatrix}$$

$$\begin{array}{ccccc} C & \hookrightarrow & G & \twoheadrightarrow & \overline{G} \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{F}_2^g & \hookrightarrow & W_{2g} & \twoheadrightarrow & S_g \end{array}$$

Proposition (Let $\overline{A} = (A_1, \dots, A_g)$)

$$\prod \beta_i^{A_i} = 1 \Leftrightarrow M \cdot \overline{A} = 0$$

Corollary

$$\delta = \text{rank } M = \dim_{\mathbb{Q}} V.$$

Tate conjecture

1 **Tate Conjecture:** $H^{2j}(A)(q^j)$ is algebraic

2 **Theorem:** $H^2(A)(q)$ is algebraic

1 $H^i(A) = \wedge^i H^1(A)$

1 If $H^{2j}(A)(q^j) = \wedge^j H^2(A)(q)$, then the Tate Conjecture is true.

2 This can fail, and such a class in $H^{2j}(A)(q^j)$ which does not come from degree 2 is called **exceptional**.

1 $\delta = g$ implies that there are no exceptional classes.

2 Idea: if $\alpha_i^2 \alpha_j^2 = q^2$, then there is an exceptional class in the image of

$$H^2(A)(\alpha_i^2) \wedge H^2(A)(\alpha_j^2) \rightarrow H^4(A)(q^2)$$

(see Zarhin).

Lemma: one can uniquely partition

$$\{1, \dots, g\} = \bigsqcup_{i=1}^m T_i \text{ such that}$$

$\sigma \in C$ is constant on each T_i and each T_i is maximal for this property.

Write:

$$\mathbb{Q}^g = \bigoplus_{i=1}^m \mathbb{Q}^{\oplus T_i}$$

Corollary: suppose that $g > 1$ and set $g' = g/m$.

Then $\delta = im$ for some $i \in \{1, \dots, g'\}$; $\delta \geq m$.

When $\overline{G}$ is not primitive, analyze $H_i = \text{Stab}_{\overline{G}} T_i$

Theorem

- 1 If g/m is prime, then $\delta \in \{m, g - m, g\}$.
- 2 If H_i acts 2-transitively on T_i , then $\delta \in \{g, g - m\}$.